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# Series arc fault identification method based on wavelet transform and feature values decomposition fusion DNN

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ABSTRACT

In low voltage distribution systems, AC series arcing is highly random and the fault characteristics are influenced by the type of loads. As the variety of loads connected to the system increases, a standard for uniform detection between varieties of loads is difficult to find. For such issues, a neural network algorithm based on wavelet analysis and feature value decomposition is proposed. Firstly, the DB5 wavelet base is used to decompose the collected data into 5 layers of wavelet and the wavelet coefficients of the first layer are extracted for the construction of the Hankel matrix. The first feature extraction and data compression are completed. The matrix is decomposed by Eigenvalue Decomposition (EVD) and used to construct the eigenvector $\sigma$, and the second feature extraction and data compression are completed. The mean value $d_1$, the root mean square value $d_2$ and standard deviation value $d_3$ of $\sigma$ are extracted, the third feature extraction and data compression are completed. The three values are taken as input to the neural network to train the Deep Neural Network (DNN) fault detection model, and the compressed input scale is only $1 \times 3$. The DNN fault detection model built by this paper is characterized by low complexity and high timeliness. The complexity of the model is reduced in terms of driving data and network structure, and the model can be trained in just 33 s. The model is extremely time efficient because it monitors faults in terms of current half-cycles. The experimental results show that, within the permissible range of GB14287.4–2014 standard, the model has good detection effect on both training set loads and non-training set loads, and the overall recognition rate can reach 98.7%.

## 1. Introduction

In residential places, fault arcs may be caused by the fall of sockets, aging of power strips or lines, overloading, and interface breakage. Arc faults are an important cause of electrical fires and are a form of self-sustaining discharge [1]. When the voltage is higher than 12 V and the current exceeds 0.25 A, the two live conductors are about to touch or separate. An arc will be generated in the conductor gap, which will generate a large amount of heat energy, accompanied by high-temperature residue sputtering, which can easily ignite the surrounding combustible and explosive materials, causing fire and explosion accidents, endangering people's lives and property safety [2]. However, in practical engineering applications, it is difficult to accurately detect it due to its characteristics of strong concealment, large randomness, and great influence by load [3,4].

At present, the research progress of arc mathematical model used for fault arc detection is very slow, and more stay in the stage of theoretical research and experimental simulation. A new arc model was formed by combining a Cassie model and two Mayr models to evaluate the DC breaking capacity of molded case circuit breakers [5]. A Cassie-Mayr combination model is constructed through the transition function, and the improved arc model is simulated and verified, which can describe the dynamic characteristics of the arc more accurately [6]. The arc sound, arc light, temperature, pressure and other physical phenomena accompanying the arc combustion are usually used as the basis for fault diagnosis. The bimodality of electromagnetic and acoustic waves is used for arc fault detection and localization, which can detect arc faults in the monitored area [7]. The modulo maximum value of the electromagnetic radiation signal of the arc fault is extracted to obtain the eigenvalue, and a reasonable threshold is set according to the eigenvalue to judge whether an arc fault occurs [8]. Physical features are used for detection requiring sensor positions to be fixed, and the randomness of the fault arc causes the position to be uncertain. If the sensor installation position is far away from the location of the fault, the detection effect will be

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greatly reduced. The method for identifying arcs based on changes in the circuit voltage signal characteristics when an arc fault occurs in [9]. The identification method of voltage signal characteristic change is hindered in practical application due to the large change of load position range and the difficulty of sampling effective voltage. The methods for identifying changes in current signal characteristics include: Methods for fault identification are always based on one or some characteristics of the series arc current, such as time-domain and frequency-domain characteristics [10–12], Fourier transform [13,14], wavelet transform analysis[15–18], etc. In engineering practice, the model is greatly affected by [19]. According to the zero-break phenomenon that the arc is extinguished near the current zero-crossing point when an arc fault occurs, the two-dimensional characteristic quantity established by the information dimension and the zero-rest is used as the characteristic parameter to identify the arc fault [20]. When the non-linear load is working normally, there is a similar zero-stop phenomenon in the current, which will also lead to misjudgment. Wavelet transform has good time-frequency localization characteristics, and has the ability to characterize local characteristics of signals in both time and frequency domains. It has more advantages than other methods in dealing with mutated signals and non-stationary signals. At present, wavelet transform has been widely studied for series arc fault detection[15–18]. The wavelet entropy value is used to quantitatively describe the energy distribution characteristics of the current signal in the time-frequency domain, so as to realize the extraction of the series arc fault characteristics [17]. Morphological filtering combined with the fourth-scale wavelet trans-form function is used to realize the identification of load normal state and arc fault state [18]. With the continuous deepening of wavelet transform in the field of series arc fault detection, it also shows some shortcomings: the simple use of wavelet transform is not obvious for the distinction between series arc fault and normal state. One is that there is a lot of redundant information in the transformed feature signal. Second, there is a characteristic overlap between normal and fault signals between different load types. The above shortcomings increase the difficulty of analysis and interpretation of wavelet transform results.

With the rise of artificial intelligence, various neural networks and machine learning algorithms have been widely used in the field of arc faults. There are certain limitations in the number and reliability of the applicable load types by using the neural network method alone. The total number of parameters and training samples increases with the number of load types. This leads to high computational cost and poor robustness of the classifier, and poor recognition of non-training set loads that have not participated in training.

Therefore, in order to satisfy the fault detection among various load types, it is necessary to perform data preprocessing and feature extraction. The data dimension can be effectively compressed and the fault features are highlighted. The feature set is used as the driving data of the neural network, which can greatly reduce the computational pressure of the classifier, so the training is completed quickly, and the computational cost is greatly reduced. The combination of empirical wavelet transform, singular value decomposition and support vector machine is used in [21]. A recognition method based on kernel principal component analysis and firefly algorithm optimization support vector machine is proposed. The kurtosis and skewness of the fifth and sixth principal components were used as arc fault features using kernel-based principal component analysis. Recognition method combined with support vector machine is proposed in[22]. The fundamental wave, the third harmonic, the fifth harmonic, the stagnation at zero-crossing, and the integral of the time-domain waveform are used as characteristic quantities, which are combined with the neural network [23]. In [21–23], there are problems of few types of training set loads and insufficient generalization ability to non-training set loads. Analyze and extract different aspects of arc features through time domain, frequency domain and wavelet packet energy, and use multi-feature fusion to train the arc fault detection model [24–26]. In the multi-feature fusion algorithm, the weight distribution between each feature is a complex problem.

In summary, at this stage in the field of low voltage AC arc fault detection, the detection accuracy is the primary of evaluation metric for fault detection algorithms, but not the only. The ease of implementation in hardware and timeliness of detection are also important. The algorithm should be simple in structure, small in computation, time efficient and easy to implement in hardware in order to make the fault detection algorithm more practical in engineering. Therefore, a series arc fault identification method based on based on wavelet transform and feature values decomposition fusion DNN is proposed. The specific contributions are as follows:

(1) The dimensionality reduction method by EVD is adopted to decompose the wavelet coefficients after wavelet transformation. Three evaluation indexes based on eigenvector were established and as the input of the DNN, which not only retains the unique advantages of wavelet transform in arc fault detection, but also reduces the dimension and redundancy of wavelet coefficients as input.
(2) An arc fault detection model based on DNN is established. The proposed model can accurately perform fault identification only at a sampling rate of 5 kHz, the final drive data scale is only $1 \times 3$, and the number of hidden layers in a neural network is only three, which greatly reduces the training time and detection time, and the calculation cost is reduced.
(3) An arc fault detection algorithm based on wavelet transform and EVD fusion DNN is proposed. Traditional algorithms and neural networks are well combined. Compared with other methods in terms of recognizetion accuracy, identification time, driving data scale, and applicable load range, the method in this paper has higher comprehensive performance.

## 2. Data acquisition and arc characteristic analysis

### 2.1. Data collection

The low-voltage AC series arc fault test platform built according to the GB/14,287.4–2014 standard is shown in Fig. 1. It is aimed at the civil voltage level, 220 V/50 Hz and the maximum power level does not exceed 10 kW. It consists of AC power supply, air switch, current probe, electrical load, arc generator and oscilloscope device.

Refer to the commonly used typical loads in the GB/14,287.4 standard for data collection, as shown in Table 1.

### 2.2. Analysis of current waveform characteristics

First, digitally filter the collected current signal. The filtering principle is to replace the value of a point in the digital sequence with the average value of each point value in a neighborhood of the point. It is a commonly used nonlinear smoothing filter. To reduce interference and noise, the filtered current frequency is 5 kHz. The current waveforms of several typical loads after filtering were shown in Fig. 2.

In Fig. 2, $t_{arc}$ is the time when the fault occurs. The electric heater is a resistive load, and its electrical characteristics before and after arcing are the same as those of the resistive load. In the initial small arc area, it is similar to its own normal current. There is a "zero rest" phenomenon after appear arc fault, similar loads are also microwave oven + resistance, vacuum cleaner, computer + microwave oven, etc. For electric drills and computers, due to the different operating conditions and working principles, the circuit presents different characteristics during normal operation. When the load is working normally, the current also has the characteristics of zero rest. When a fault arc occurs in the line, the fault current will lose its periodicity, and there will be half-wave loss and high harmonic content.

Affected by the reason of arc fault, load capacity, load nature, etc., it only depends on a certain parameter in the time domain or frequency domain before and after the arc as the basis for fault arc detection.

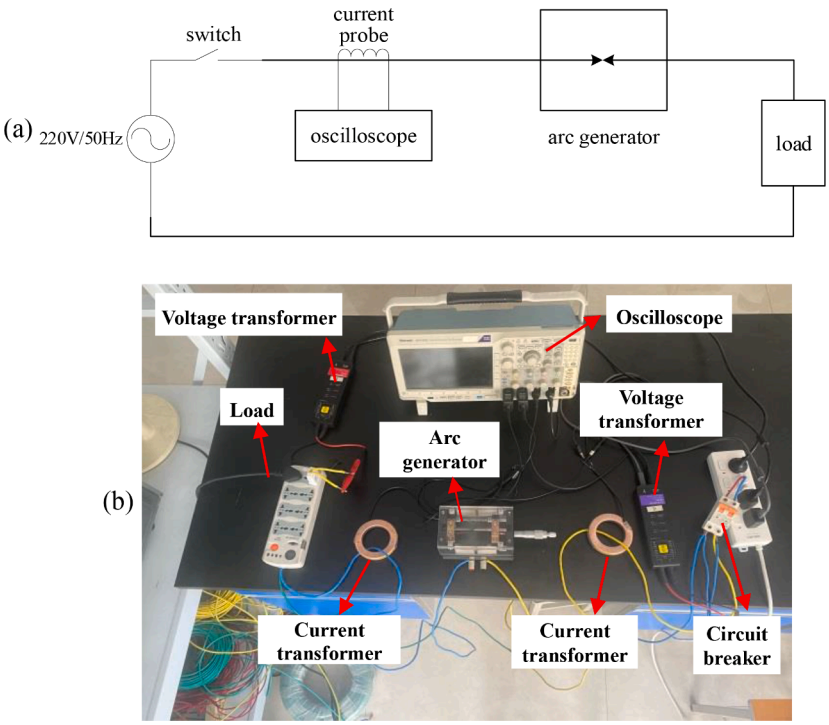



**Fig. 1.** AC series arc fault test platform. (a) Experimental platform circuit diagram (b) Physical map of experimental platform.

**Table 1**
Types of load in experiment.

| Loads | Load type | Load name and parameters(220 V) |
|---|---|---|
| Linear load | Resistive load | Resistor 1000 w |
| Linear load | Resistive load | Electric heater 800 w |
| Nonlinear load | Capacitor start motor | Electric drill 2500 w |
| Nonlinear load | Vortex | Induction cooker 2200 w |
| Nonlinear load | Switching power supply | Computer 600 w |
| Nonlinear load | Magnetron | Microwave oven 800 w |
| Nonlinear load | Series motor | Vacuum cleaner 1000 w |
| Nonlinear load | Combined load | Microwaveoven800 w+resistance800 w |
| Nonlinear load | Combined load | Computer 200 w+vacuum cleaner 1 kw |
| Nonlinear load | Combined load | Microwave oven 600 w+computer600 w |

Although fault detection can be achieved, it is difficult to find common thresholds are prone to misjudgment, especially in the initial small arc stage, the time domain signal and frequency domain signal before and after arcing are less distinguishable, and the cross-criteria value is prone to occur.

## 3. Feature extraction of series arc fault

### 3.1. The analysis of discrete wavelet transform

The "zero rest" feature alone is not suitable for non-linear loads as a detection method, and further processing is required. Wavelet transform is widely used in featureity detection, which is very suitable for the processing of "zero rest" characteristics. It is necessary to introduce discrete wavelet transform (DWT) to reduce redundancy and improve calculation speed without losing information in the signal [27].

The continuous wavelet transform (CWT) of a continuous signal $x(t)$ is defined as:

$$wt(s,\tau)=\frac{1}{\sqrt{s}}\int_{-\infty}^{+\infty}x(t)\varphi\left(\frac{t-\tau}{s}\right)dt \tag{1}$$

The $\varphi$ is the base wavelet, $s$ is called the scale parameter, stretch or stretch the base wavelet. $\tau$ is called the translation parameter, determine the time position at which to analyze the signal $x(t)$.

The specific method of discretization is to discretize the scale parameter and the translation parameter:

$$\begin{cases} s=s_0^j \\ \tau=k\tau_0 s_0^j \end{cases} \quad s_0>1,\tau_0\neq 0,j\in\mathbf{Z},k\in\mathbf{Z} \tag{2}$$

The corresponding wavelet family is expressed as:

$$\varphi_{j,k}(t)=\frac{1}{\sqrt{s_0^j}}\varphi\left(\frac{t-k\tau_0 s_0^j}{s_0^j}\right) \tag{3}$$

The wavelet transform of the signal is:

$$wt(j,k)=\langle x(t),\varphi_{j,k}(t)\rangle=\frac{1}{\sqrt{s_0^j}}\int_{-\infty}^{+\infty}x(t)\varphi\left(\frac{t-k\tau_0 s_0^j}{s_0^j}\right)dt \tag{4}$$

The most common form of discrete wavelet transform is the scale parameter $s_0$=2, translation parameter $\tau_0=1$, this discretization form makes its wavelet function family constitute an orthonormal basis.

After the DB5 wavelet base is used to decompose the 5 loads of electric heater, vacuum cleaner, computer + microwave oven, computer and induction cooker in five layers, the first layer of wavelet coefficients are extracted and their absolute values are taken to obtain the parameter $D$, as shown in Fig. 3.

In Fig. 3, the fault occurs at 0.1 s. There is a large variation in the wavelet coefficients in the first layer before and after the fault arc occurs, but the values are crossed over in the five loads. When the heater is working normally, $D$ is between 0~0.001, and it is between 0.003~0.02 when it fault. Under the same state, for computer and electric heater, their wavelet coefficient value of the first layer differs by about 100 times of. In general, the wavelet coefficient value of the first layer is small, which is not conducive to fault analysis. It is difficult to use wavelet transform alone to detect arc faults between various loads.

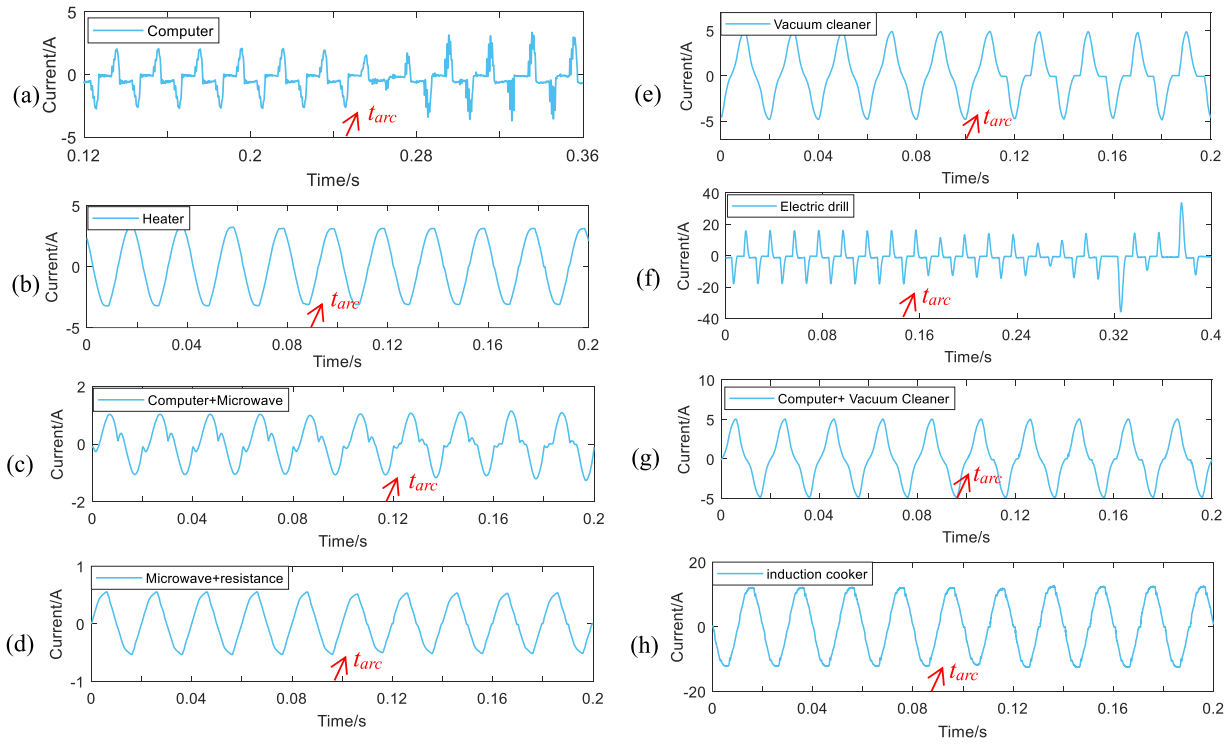



**Fig. 2.** Typical load normal and fault current waveform. (a) computer. (b) Heater. (c) Computer + Microwave. (d) Microwave + resistance. (e) Vacuum cleaner. (f) Electric drill. (g) Computer + Vacuum Cleaner. (h) Induction cooker.

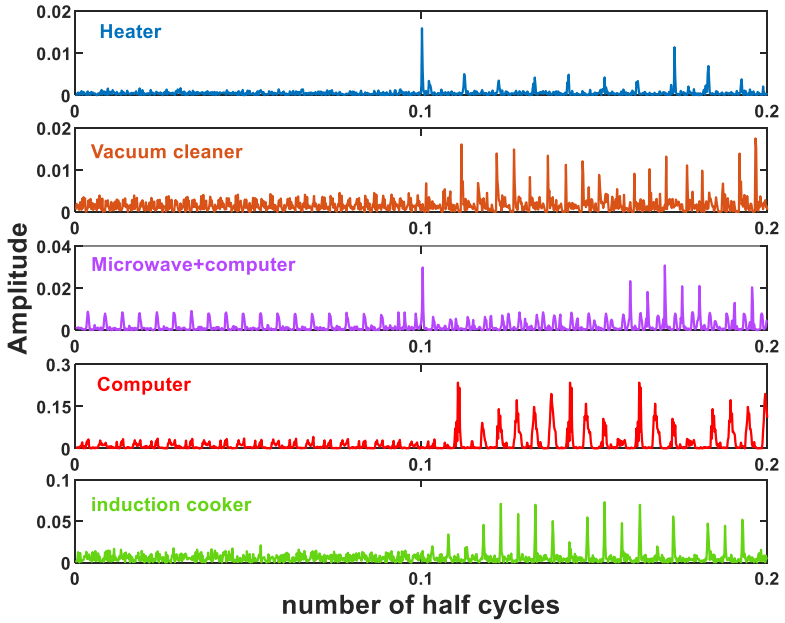



**Fig. 3.** The first-level detail components after wavelet transform.

### 3.2. The basic principle and fault analysis of EVD

Suppose for a real matrix$A \in R^{m\times n}$, there must be an orthogonal matrix $V \in R^{m\times n}$ make Eq. (5) true [28]:

$$A = VSV^{-1} \tag{5}$$

In Eq. (5):

$$V = [v_1, v_2, v_3 \cdots v_m] \in R^{n\times n} \tag{6}$$

$$S = [\mathrm{diag}(\sigma_1, \sigma_2, \sigma_3 \cdots \sigma_r), 0] \tag{7}$$

The$A = \min(m, n)$,$\sigma_1 > \sigma_2 > \sigma_3 \cdots > \sigma_r > 0$, the 0 is a zero matrix,$\sigma_i (i = 1 \cdots r)$called the feature values of the matrix $A$, Eq. (7) is called the feature value decomposition of $A$.

Constructing a suitable matrix $A$ is the key to signal processing by feature value decomposition. The Hankel matrix is constructed according to the signal sampling sequence. After the Hankel matrix is decomposed by EVD, a multi-layer component signal is generated for the signal. For a sampled sequence vector of $N$ points:$X = (x_1, x_2, x_3 \cdots x_N)$. It

is used to construct the Hankel matrix method as follows:

1) Select a subsequence$(x_1, x_2, x_3 \cdots x_n)$of length n from the signal time series as the first row vector of the matrix.
2) Move one step to the right and extract$(x_2, x_3, x_4 \cdots x_{n+1})$as the second row vector of the matrix, and so on, a total of *N- n* + 1 row vectors can be obtained.
3) These row vectors obtained above are formed into a matrix as shown in Eq. (8), and this matrix is called a Hankel matrix.

$$A = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \\ x_2 & x_3 & \cdots & x_{n+1} \\ \vdots & \vdots & \vdots & \vdots \\ x_{N-n+1} & x_{N-n+2} & \cdots & x_N \end{bmatrix} \tag{8}$$

The specific meaning of each element in the corresponding formula of EVD: The matrix *A* represents the feature matrix constructed by the discrete wavelet coefficient sequence *D*, The feature values$\sigma_1, \sigma_2, \cdots, \sigma_r$, represents the size of the characteristic quantity of the current signal collected in this time band, and is uniquely determined by the matrix *A*. If these non-zero feature values are composed of eigenvectors$\sigma_1, \sigma_2, \cdots, \sigma_r$, the eigenvector uniquely characterizes the characteristics of the current signal. The Fig. 4 is a comparison diagram of the characteristic vector $\sigma$ of different load faults and normal signals.

It can be seen from Fig. 4 that after the feature value decomposition of the first layer wavelet coefficients, the fault state is quite different from the eigenvector in the normal state. Through EVD, the high-dimensional feature matrix *A* is compressed into a low-dimensional outlier vector reflecting the essential characteristics of matrix *A*, compared with the 54-dimension of the discrete wavelet coefficient sequence *D*, the dimension of the feature value vector $\sigma$ is greatly reduced to 27-dimensions. In the $\sigma$, the characteristic curves of the first 15 characteristic points before and after the fault vary greatly and cover more characteristic information, which is called the sensitive area. Then the characteristic curves of the last 12 characteristic points almost overlap before and after the fault, covering a very small amount of characteristics, which is called the non-sensitive area. This is also another feature after eigenvalue decomposition. In the eigenvalue sequence, the first 50% of the eigenvalue sequence will contain more than 90% of the feature information. It is widely used in image processing and can greatly compress the data dimension [29]. Therefore, the first 15 feature points are finally selected.

### 3.3. The feature quantity is extracted

In order to more intuitively reflect and evaluate the performance of a segment of the signal, there are three characteristic parameters about the average value d1 and the root mean square value d2 and standard deviation d3 are calculated. The expressions of d1, d2 and d3 are:

$$d_1 = \frac{\sigma_1 + \sigma_2 + \cdots + \sigma_r}{r} \tag{9}$$

$$d_2 = \sqrt{\frac{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2}{r}} \tag{10}$$

$$d_3 = \sqrt{\frac{1}{r-1}\sum_{i=1}^{r}(\sigma_i - \overline{\sigma})^2} \tag{11}$$

The *r* is the number of eigenquantities in $\sigma$, which are 15. GB/14,287.4 stipulates that if 14 or more half-cycle fault arcs occur within 1 s, the Arcing Fault Detectors (AFD) shall issue an alarm signal within 30 s, therefore, fault analysis should be performed in units of current half-cycles. The characteristic parameters $d_1$, $d_2$ and $d_3$ are obtained for the five loads of electric heater, vacuum cleaner, computer + microwave oven, computer and induction cooker. The fault occurred at the 21st half cycle, the first 20 half cycles are when the load is in normal operation, and the last 20 are in the state of arc fault. The results are shown in Fig. 5.

As shown in Fig.5, it can be seen that before and after the arc fault occurs, the three evaluation indicators have great changes, but they cannot be used as the final detection basis. It can be used as the driving data of neural network for fault detection. The advantage of using it as driving data is that the characteristic information in the data is obvious and the data amount is small, which can effectively reduce the complexity of neural network.

## 4. DNN structure and establishment of fault detection model

### 4.1. The establishment of DNN fault detection model

The neural network structure applied to arc fault detection is shown in Fig. 6, which is composed of input layer, hidden layer and output layer. The feature vector *d* is the input of the network, hidden layer is

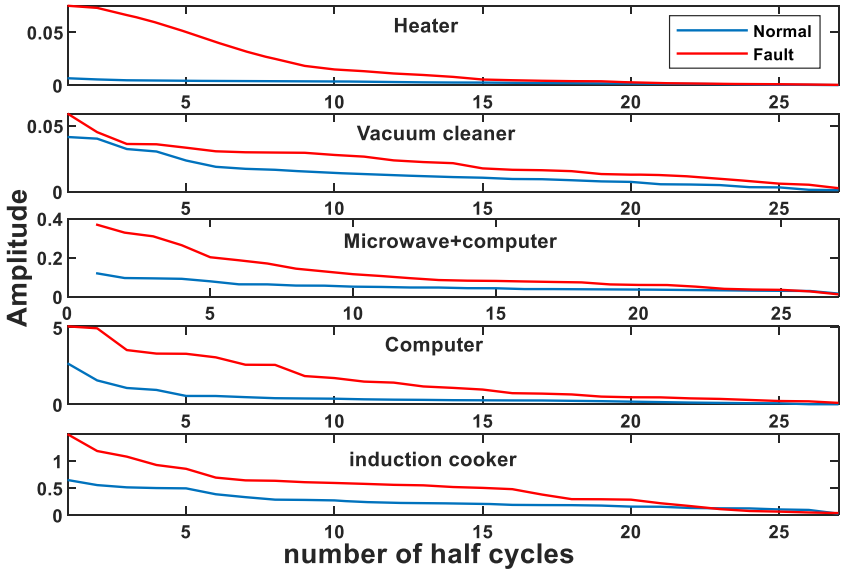



**Fig. 4.** Comparison of feature values in different states of load current half cycle.

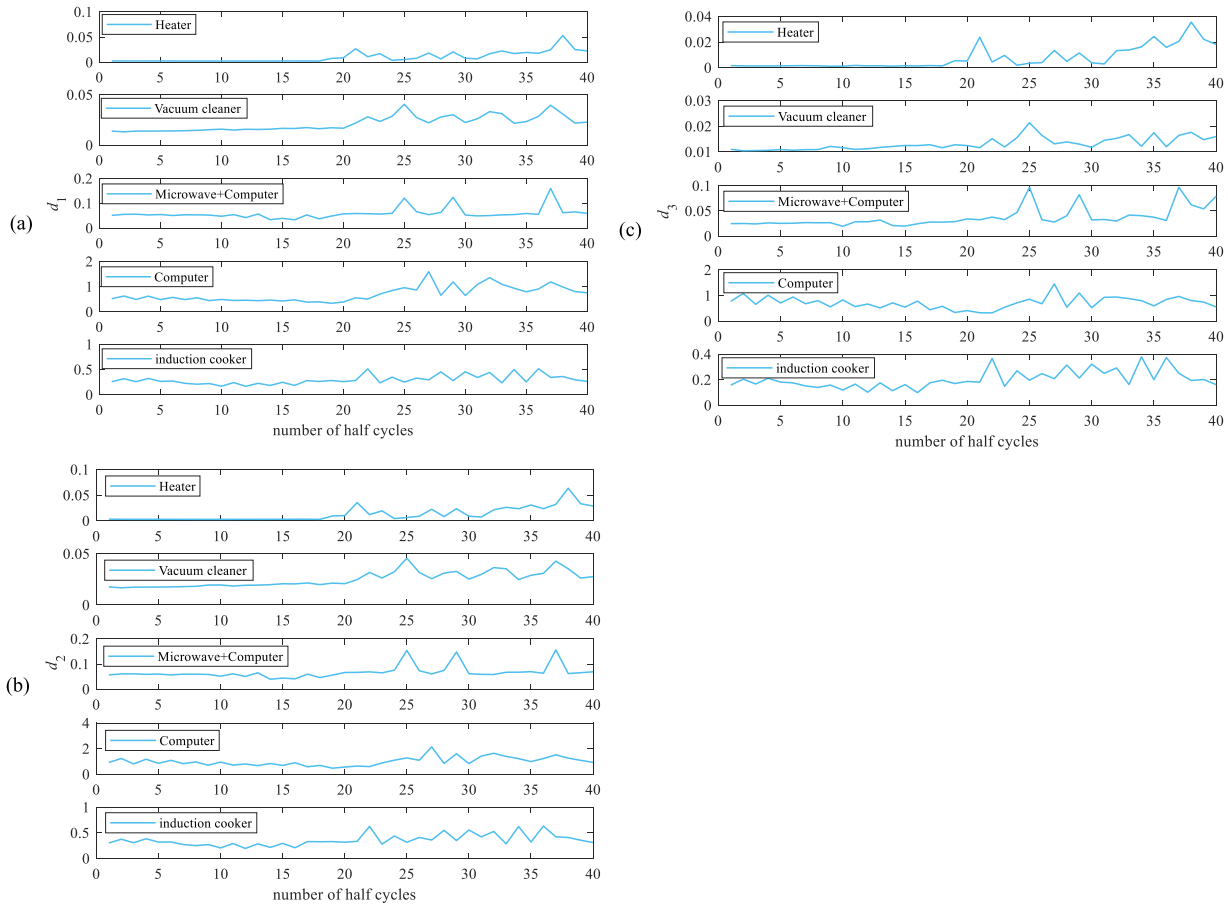



**Fig. 5.** The values of three characteristic parameters of different loads. (a) The $d1$ value between different loads (b) The $d2$ value between different loads (c) The $d3$ value between different loads.

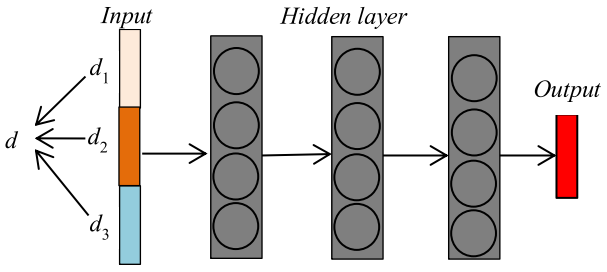



**Fig. 6.** The structure of the DNN fault detection model.

composed of neurons with Gaussian activation. The output layer uses the sigmoid function as the activation function for binary classification. The network structure is simple and the computational complexity is reduced.

As shown in Fig. 6, the hidden layer of the network contains three layers, which is precisely a type of deep network and is superior to shallow perceptrons in terms of performance. The number of nodes of neurons in the hidden layer was selected according to the empirical formula mentioned in reference [24], and finally, the exact number of neurons in the 3-layer hidden layer was determined as 16+32+16.

The specific steps of extracting current signal features based on the combination of discrete wavelet transform and EVD are as follows:

(1) The collected current signal is digitally filtered by mean filter, the frequency after filtering is 5 kHz, and the collected current signal per half power frequency cycle (100 values in total) is decomposed into 5 layers by the wavelet basis function DB5, and the discrete value is extracted. Wavelet coefficient sequence $D$ (54 values in total), and construct Hankel characteristic matrix $A$ for $D$.

(2) Perform EVD on matrix $A$ to get eigenvectors $\sigma$ (15 values in total).

(3) According to equation (9–11), $\sigma$ is calculated by the eigenquantities $d_1$, $d_2$ and $d_3$ to form the eigenvector $d$.

Through the above steps, the wavelet coefficient sequence $D$ and $\sigma$ are extracted from the current data collected by 10 kinds of loads, and the eigenvector d is obtained. Based on the folder where the data file is located, there is an arc fault. Marked as 1, no fault is 0. A total of 3300 sets of training data are formed, of which 165 sets are normal and 165 sets are faulty for each load.

For a more intuitive representation of the input and output feature matrices, a section of vacuum cleaner load current data was captured, as shown in Fig.7. The feature quantities were also extracted to generate the input and output feature matrices, as shown in Table 2.

The data is read in rows, with each row being a set of training data. The first 20 sets are the input and output quantities in the normal state and the last 20 sets are the input and output quantities in the fault state.

Input the preprocessed 3300 groups of data into the DNN model, and 3000 groups as the training set (1500 normal, 1500 fault), and 300 groups as the test set(150 normal, 150 fault) . Passed the test, it is found that the learning rate is set to 0.001, the training effect is better. The training effect of the model is shown in Fig. 8, and the total training time is 33 s. After 100 iterations, the training accuracy rate is stable at around 98%. At the same time, the loss value curve of the network also tends to

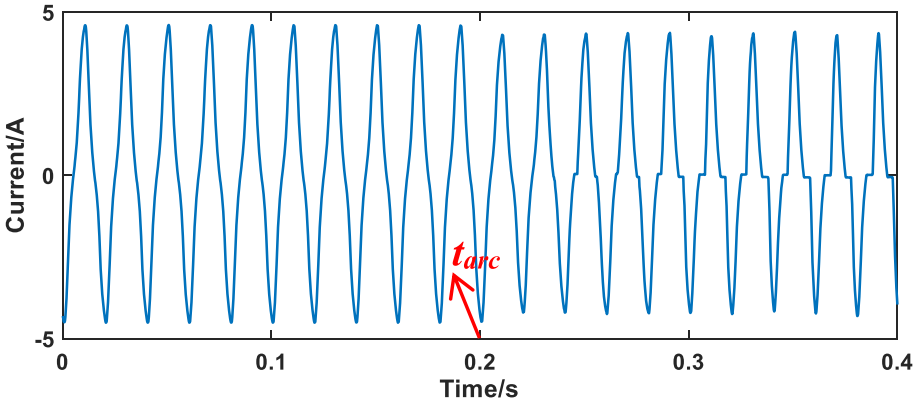



**Fig. 7.** Current waveform diagram for vacuum cleaner.

**Table 2**
Inputs and outputs feature matrix.

| Number of rows of the matrix | Inputs | | | Outputs |
|---|---|---|---|---|
| 1 | 0.0139 | 0.0176 | 0.0109 | 0 |
| 2 | 0.0132 | 0.0167 | 0.0104 | 0 |
| 3 | 0.0139 | 0.0173 | 0.0105 | 0 |
| … | … | | | … |
| 19 | 0.0172 | 0.0213 | 0.0127 | 0 |
| 20 | 0.0168 | 0.0207 | 0.0124 | 0 |
| 21 | 0.0219 | 0.0247 | 0.0116 | 1 |
| 22 | 0.0281 | 0.0317 | 0.0151 | 1 |
| 23 | 0.0235 | 0.0262 | 0.0119 | 1 |
| … | … | | | … |
| 38 | 0.0308 | 0.0353 | 0.0176 | 1 |
| 39 | 0.0219 | 0.0262 | 0.0147 | 1 |
| 40 | 0.0227 | 0.0276 | 0.0159 | 1 |

be stable, and the model begins to converge.

The output result of DNN model is a binary classification problem, so when the output result is greater than 0.5, the output result is 1, which is judged as half cycle of arc fault. When the output is less than 0.5, it is judged to be a normal half-cycle. For part of the verification set, the comparison between output results, prediction results and real labels is shown in Fig. 8 (b). The test results of the verification set are shown in Fig. 8 (c). The recognition results of 150 groups of normal half-cycle and fault half-cycle are 149 and 147, respectively.

In order to further verify the degree of adaptation between the feature vector $\sigma$ and the network, cross-validation is used to test the performance of the model. *Re*-collect and generate 6000 sets of training data for 10 typical loads, randomly select 3000 as the training set and 300 as the test set for training, repeat 50 times, count the test results on the test set. After 50 cross-validations, the range of accurately identified 150 sets of normal current half-cycle data ranges from 146 to 150, and the average recognition rate is 98.6%. The range of accurately identified 150 sets of fault current half-cycle data ranges from 142 to 150, with an average recognition rate of 97.3%.

For neural networks applied to 1-dimensional data processing, single-layer ANN, two-layer ANN, 4-layer DNN and 5-layer DNN were constructed for comparison with the 3-layer DNN used in this paper in order to test the effect of network depth on detection results. In which, the data sets are used to be the 3000 sets of training set and 300 sets of test set established by this paper, the detection results for the test set are shown in Fig. 9.

As shown in Fig. 9, for both faulty and normal half-cycles, when the number of the hidden layer is greater than or equal to three, the model is significantly more accurate in identifying the test set than when the hidden layers are a single layer or two layers. However, when the hidden layer reaches 3 layers, the accuracy of testing the test set no longer improves with increasing network depth. At this point it becomes more important to adjust the hyperparameters in the model, such as the number of nodes in each hidden layer, and the learning rate. The reference [24] similarly demonstrates that DNN outperform shallow ANN in terms of performance.

### 4.2. Fault detection algorithm flow

GB/14,287.4 standard stipulates that if 14 and more half-cycle fault arcs occur within 1 s, the fault arc breaker should issue an alarm signal within 30 s. In accordance with the above standard requirements, current data is collected in units of each half-cycle.

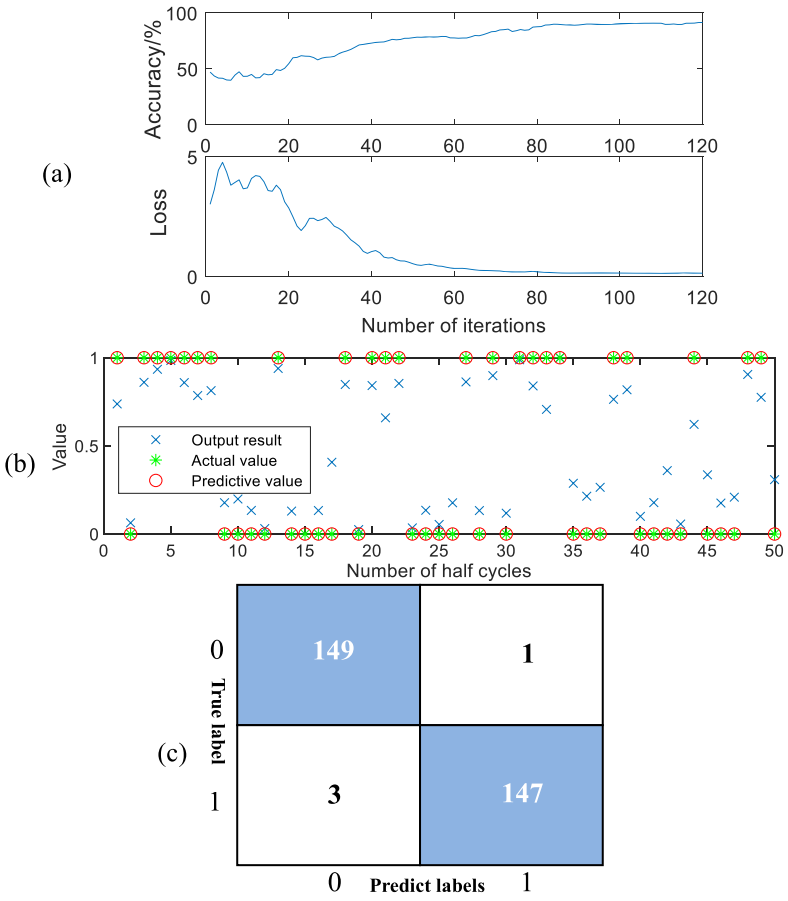
As shown in Fig.10, firstly, using the method proposed in this paper, the feature matrix $d$ is extracted from the collected half-cycle current data and fed into the DNN fault detection model. If the model prediction result is greater than 0.5, the fault half-cycle is judged, and if the model prediction result is less than or equal to 0.5, the normal half-cycle is judged. Once the discrimination is complete, the current data is updated. If the number of fault half-cycles reaches 14 within 1 s (100 current half-cycles), the arc fault is determined to have occurred and the fault detection is complete. When the detection time limit reaches 1 s, but the number of fault half-cycles has not reached 14, the accumulated results are cleared and a new round of accumulation takes place.

## 5. Verification and comparative analysis of DNN fault detection model

### 5.1. DNN fault detection model performance detection

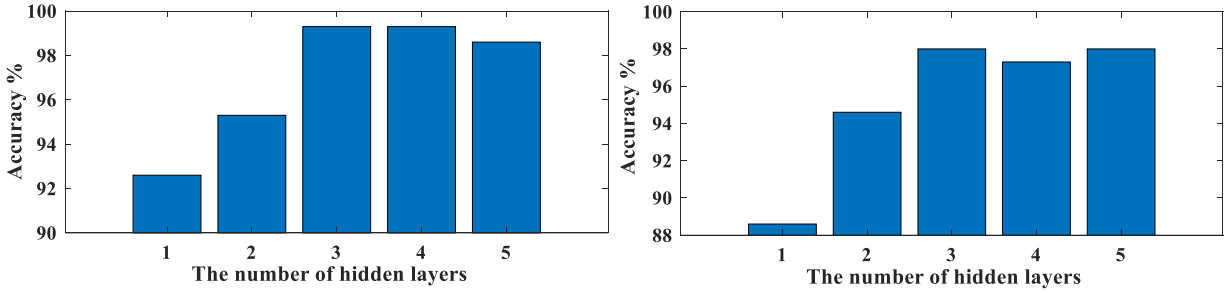
In order to further verify the reliability of the DNN model for fault detection of the load types in the training set, 100 sets of effective current data were collected for each of the 10 typical loads participating in the model training, totaling 1000 sets. Each set of data contains 100 normal half-cycles and 100 fault half-cycles (2 s of current data, 1 s as normal, and 1 s as fault). The arc fault detection performance of the DNN model is verified in MATLAB, and the detection results are shown in Table 3.

The results show that in the detection of 100 arc fault half cycles (1 s) of fault data, except for 4 of the 50 sets of tests on microwave oven loads, the recognition effect did not reach the standard of 14. And the other 50 sets of tests for each load were completed efficient identification of 14 and more arc half cycles. Models with excellent detection performance, such as those recently proposed by reference [23], reference [30] and reference [31], have not been tested on microwave ovens. The microwave oven is a "shielded" load and because it contains a large number of functional circuits, the characteristic information in the fault current signal is absorbed and the fault characteristic information is weak, so there are four missed detections in the test results. During normal operation of the load, the misjudgment of 100 non-fault arc half cycles




(a) Training process (b)Partial verification set output results, real results, and predicted results (c)Detection results on the test set.

**Fig. 8.** Training effect of DNN model driven by *d*.




(a) Test results for the normal half cycle of the test set (b) Test results for half cycles of fault in the test set

**Fig. 9.** Detection results of the model on the test set when the hidden layer is changed.

for each load does not exceed the standard of 14, and will not cause misjudgment. In general, in 100 arc half cycles (1 s) when 10 kinds of loads fault, the overall identification accuracy of the arc half cycle is about 80%, and the accuracy of arc fault identification is 99.6%. The normal working signal produces misjudgment. The fault detection method using the model can effectively complete the status classification of 10 typical loads and realize arc fault identification.

In actual work, the types of civil electrical loads are complex and changeable, and there are often unknown loads. In order to further test the detection performance and universality of the model, 50 sets of effective current data for each of 5 load types that did not participate in model training were collected to detect non-training load types. Each set of data contained 100 normal half-cycles and 100 fault half-cycles are input into the model for detection, and the effects are shown in Table 4.

The halogen lamp is a load that has a significant impact on the arc fault detection algorithm, and its "zero rest" characteristics in the normal state are stronger than the "zero rest" characteristics of some loads in the fault state. Therefore, many arc fault interrupters on the market are strictly limited to its use. The test results for halogen lamps show that in the detection, and the accuracy can be 76% of fault identification within the GB/14,287.4 standard. The remaining four loads have good detection results and can be 100% accurate. In the detection of normal data of 5 kinds of loads, the misjudgment of the half cycle has not reached the standard of 14, and there will be no misjudgment. The overall

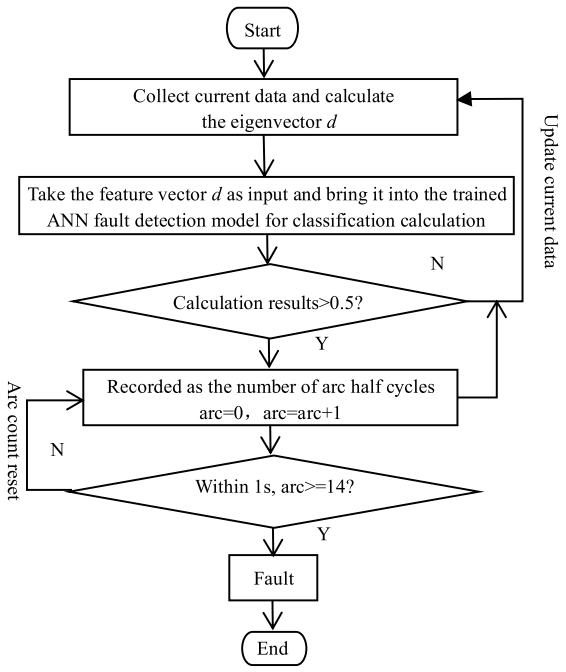



**Fig. 10.** Flowchart of fault detection for the DNN model.

**Table 3**
The DNN arc fault model on the load recognition effect of the training set.

| Load name | Correct identification number range for 50 groups of normal half-cycle data | Correct identification number range for 50 groups of fault half-cycles | Comply with GB/14,287.4,complete the number of fault identification groups |
|---|---|---|---|
| Resistor | 98~100 | 86~96 | (50/50) |
| Electric iron | 96~100 | 82~93 | (50/50) |
| Electric fan | 89~98 | 68~82 | (50/50) |
| Induction cooker | 96~100 | 72~86 | (50/50) |
| Computer | 93~100 | 85~93 | (50/50) |
| Microwave oven | 97~100 | 12~33 | (46/50) |
| Vacuum cleaner | 95~100 | 86~94 | (50/50) |
| Microwave oven+ resistance | 92~100 | 56~71 | (50/50) |
| Computer+ Vacuum Cleaner | 95~100 | 65~78 | (50/50) |
| Fluorescent lamp 36 W | 91~100 | 23~41 | (50/50) |

recognition accuracy for non-training set loads more than 90%. In general, it has high detection accuracy for non-training set loads. In practical engineering applications, the detection range of the model can be continuously optimized by continuously supplementing the types of training sets.

For 10 typical loads, when an arc fault occurs, the arc voltage values in different power frequency half cycles are not exactly the same under the same load. There is a big difference in the fault current data between different groups, but when the load is working normally, the data is relatively stable, so it is necessary to change the load power for further method verification. 50 sets of valid data for each of the 10 typical loads after the power has been changed are re-collected, of which 25 sets of data are faulty and 25 sets of non-faulty data for each load (1 s for each set of data), a total of 500 sets of data. The detection results are shown in Table 5.

The change of the overall load power has less impact on the detection performance of the DNN model, and more importantly, the impact of the characteristics of the load itself. For computer loads, misjudgments all come from the influence of control circuits such as internal rectifier circuits, which occur in the normal stage of the load. For the fluorescent lamp load, due to the small power, the signal is greatly affected by the line, equipment and environment, so there are several misjudgments.

### 5.2. Anti-jamming analysis of DNN fault detection model

Data-driven fault detection algorithms based on data are susceptible to anomaly-driven data, such as metering errors, equipment disturbances and start-up process for loads that can generate unfavorable data for detecting anomalies. Therefore, it is necessary to test the model against interference.

Fig. 11(a) shows the current data for the start-up process of the computer load. The current collection result should be 0 for 175 half cycles before the line is connected to the load. Although the power is not turned on, but the process is still being collected as smaller raw drive data. The reason is that the data comes from other disturbances in the line and is a mixture of disturbance data, containing line and equipment disturbances etc. The disturbance can be interpreted as a metering error introduced by the line and the equipment. It takes approximately 60 current half-cycles for the load to go from power on to stable operation. For the two stages mentioned above, the calculation of the three characteristic parameters of the computer start-up process was carried out and the results are shown in Fig. 11(b). The DNN model classification results for the computer start-up process are shown in Fig. 11(c). There were six false alarm cases in the 175 current half-cycles before the line was connected to the load but the power was not turned on. This error is caused by metering errors from line disturbances and equipment disturbances. During the period from when the power is switched on to the computer's stable operation, a total of 60 current half-cycles, there are eight cases of misjudgement. The cause is disturbance caused by unstable operation during load start-up. In the following 140 current half-cycles (computer load stabilization phase), there were five misjudgements. The misjudgements were caused by mixed disturbances during load stabilization. As required in GB/14,287.4, 14 or more numbers of arc half-cycles need to be detected in 1 s. Although there are a total of 19 misjudgements throughout the process, the maximum number of misjudgements is 9 within the 1 s time scale and misjudgements of the detection results are not caused. The effect of perturbations caused by the start-of the computer load can be effectively ruled out.

The process of accessing the microwave oven load during stable operation of the electric heater is shown in Fig. 12. The calculation results for the three characteristic parameters of the process are shown in Fig. 12(b). The classification results of the DNN model for the load throwing process are shown in Fig. 12(c). During the 2.5 s of stable operation of the electric heater, there were 16 instances of misjudgement. From the time the microwave oven is connected in the circuit to the stable operation of the mixed load, approximately 90 current half-cycles are passed and a total of three misjudgements exist. The misjudgements were caused by unstable operation of the microwave oven load when it was connected to the line. In the following 1.6 s, which is the smooth running phase of the load, there are 8 false positives. The misjudgements were caused by the effect of mixed disturbances on the drive data. As required in GB/14,287.4, 14 or more numbers of arc half-cycles need to be detected in 1 s. In the whole process, although there are 27 misjudgements, the maximum number of misjudgements is 8 in a time scale of 1 s, and misjudgements in arc fault detection are not caused. The effects of disturbances generated during load input and removal can be effectively excluded.

The DNN fault detection model statistics for the typical load start-up process or the process of load access circuits are shown in Table 6. Current data was collected 10 times for each type of load.

**Table 4**
DNN model load detection results on non-training.

| Load name | Correct identification number range for 50 groups of normal half-cycle data | Correct identification number range for 50 groups of fault half-cycles | Comply with GB/14,287.4 standard, complete the number of fault identification groups |
|---|---|---|---|
| Air conditioner | 91~88 | 41~62 | (50/50) |
| 1200 W Electric kettle 1500 W | 98~100 | 88~97 | (50/50) |
| Halogen lamp 55 W Electric fan 400 W+ computer 600 W | 96~100 | 8~19 | (38/50) |
| Electric fan 600 W+ electric kettle 1 kW | 93~100 | 63~76 | (50/50) |
| | 97~100 | 32~57 | (50/50) |

**Table 5**
The detection results of the DNN model that the load power is changed.

| Load name | Correct identification number range for 50 groups of normal half-cycle data | Correct identification number range for 50 groups of fault half-cycles | Comply with GB/14,287.4 standard, complete the number of fault identification groups |
|---|---|---|---|
| Resistor 1200 W | 97~100 | 86~96 | (50/50) |
| Electric iron 600 W | 95~100 | 82~93 | (50/50) |
| Electric fan 600 W | 87~96 | 68~82 | (50/50) |
| Induction cooker1800W | 94~98 | 72~86 | (50/50) |
| Computer 300 W | 83~96 | 85~93 | (46/50) |
| Microwave oven 500 W | 93~100 | 12~23 | (50/50) |
| Vacuum cleaner 400 W | 96~100 | 86~94 | (50/50) |
| Microwave oven 300 W+ resistor500 W | 90~100 | 56~71 | (50/50) |
| Computer300 W + Vacuum 400 W | 93~100 | 65~78 | (50/50) |
| Fluorescent lamp 12 W | 91~100 | 11~26 | (43/50) |

The above statistics show that, according to GB/14,287.4, 14 or more arc half-cycles need to be detected within 1 s before an arc fault is deemed to be a standard requirement. The impact of anomalous data due to metering errors, equipment disturbances, load access circuits processes etc. can be effectively eliminated by the DNN fault detection model.

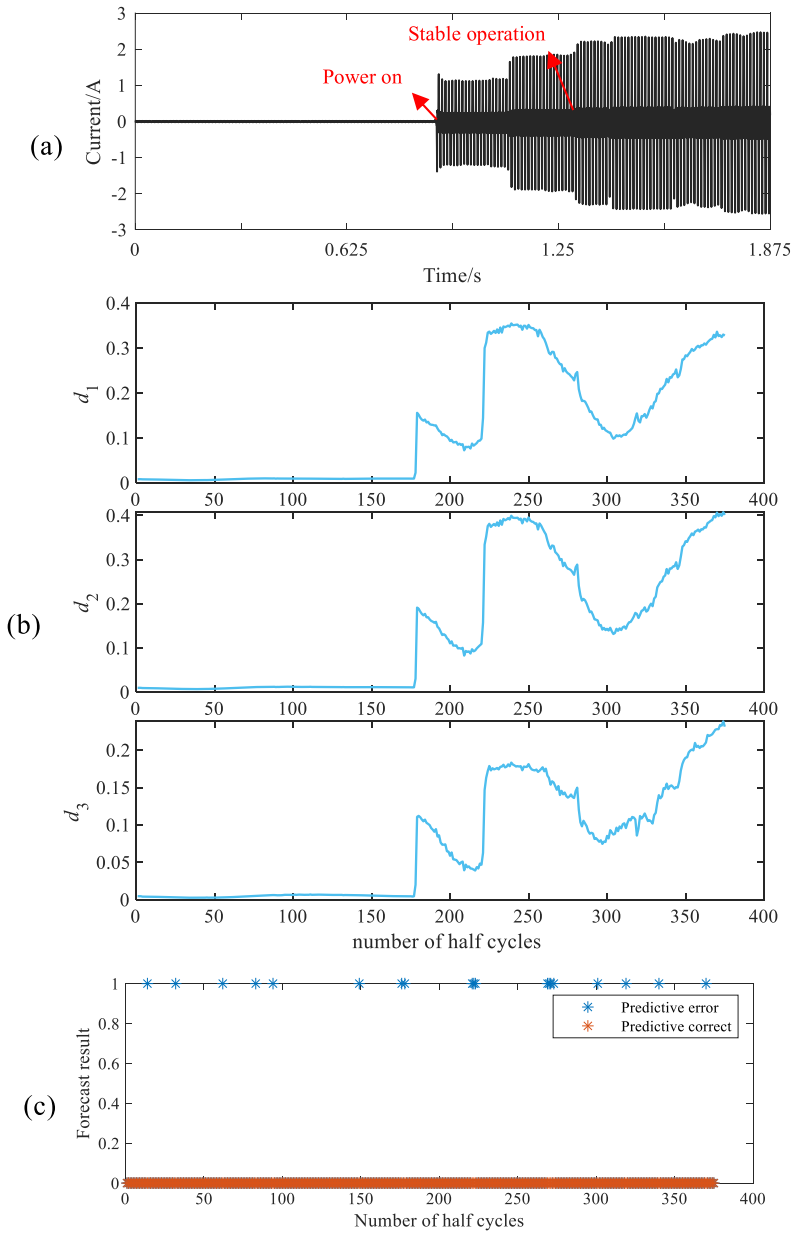
### 5.3. Method comparison

In order to verify the superiority of the proposed method, three representative methods in this field were selected for comparison with the proposed method. Both the proposed method and the selected method extract feature quantities by traditional algorithms. The feature quantity is used as the driving data of the artificial intelligence algorithm to train the arc fault detection model. The specific comparison results are shown in Table7.

The wavelet transform was likewise used as the basis of the method by reference [31] and fused with a ResNet network. Ultimately, an arc fault detection model was developed. The method is novel in introducing the field of computer vision into arc fault detection and has a high fault recognition rate. However, the input to the model is a three-channel RGB color image, which would complicate the gradient calculation. The structure of CNN is also very deep compared to DNN (applied to one dimensional data). This makes the training process and the computation of the model complex and large, and not suitable for implementation into hardware development. Therefore, this paper chooses a low sampling rate and finalizes the wavelet coefficient matrix to a simple $1 \times 3$ feature matrix and a DNN. This simplifies the training process of the fault detection model, reduces the computational complexity and thus guarantees detection timeliness. For the fault detection model proposed in this paper, the arc half-cycle can be monitored in real time and the calculation is carried out in half-cycle units. Each time a current half-cycle is elapsed, that half-cycle is discriminated and the fault is determined after a total of 14 arc half-cycles have been discriminated. The fastest fault detection time is just 14 current half-cycles, so the model has a high degree of real-time performance. Reference [31], on the other hand, collects 0.5 s of current data, which is 50 current half-cycles, and then performs fault discrimination on the entire section of current data, with a fixed detection window size and therefore relatively poor time-efficient performance. The methods such as those proposed in reference [21] and reference [24] are also built on the basis of DNN, and compress the input volume scale and model structure as much as possible during the construction of the fault detection model. However, the fault detection model proposed in this paper has the lowest sampling rate, the smallest scale of inputs and the simplest neural network structure compared to the above references.

Comparison results are summerised as follows:

(1) Through comparison, the proposed method is applicable to the most comprehensive range of load types, and all seven load types are applicable. The applicable load range is the largest and most comprehensive. In particular, the method proposed in this paper enables the identification of faults in microwave oven loads with "shielding" characteristics.
(2) The feature vector scale is the scale of the driving data, which is one of the important factors that determine the training complexity of artificial intelligence algorithms and complete rapid detection. The proposed algorithm has the smallest scale of




(a) Current generated by the computer start-up process

(b) Graph of calculated results for the 3 parameter indicators

(c)Classification results of DNN fault detection models for start-up processes

**Fig. 11.** Analysis of computer start-up process.

driving data, which is only 1 × 3. The requirements for the complexity of the artificial intelligence algorithm framework are low. It has the most convenient training process and the lowest computational cost.

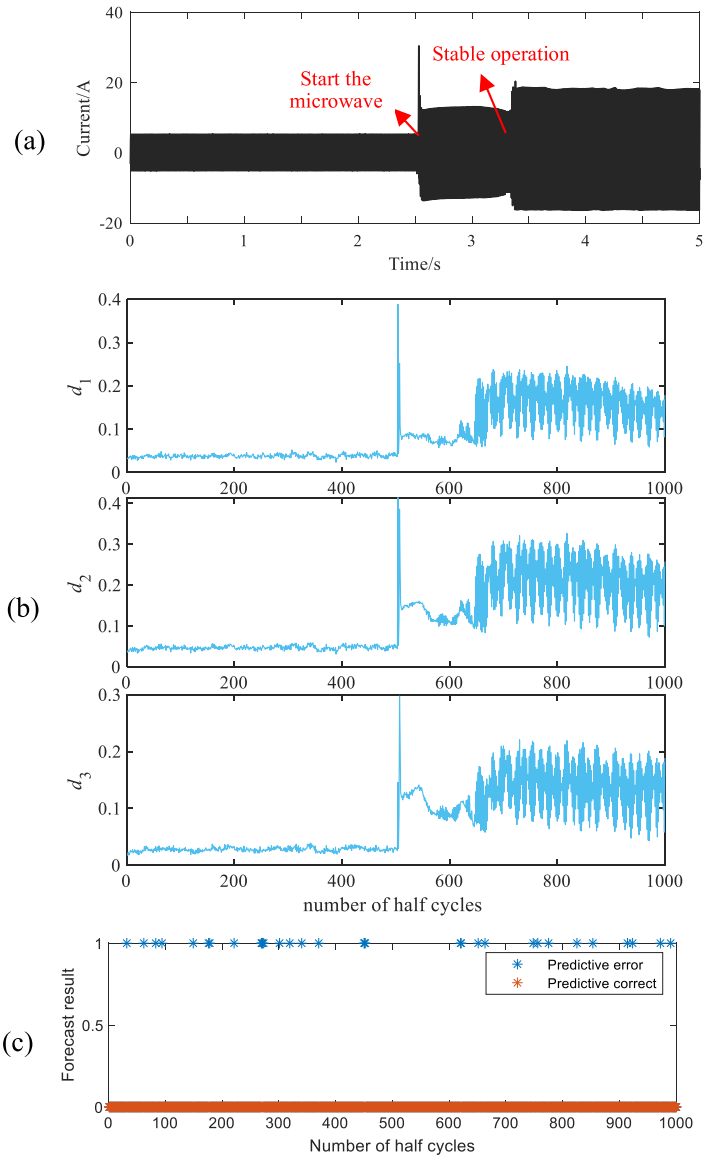
(3) The identification time represents the standard within which 14 arc half cycles should be detected as specified by GB/14,287.4. The proposed method has the shortest mean recognition time of only 0.18 s, which is the fastest detection response time.

(4) The overall recognition rate represents the average recognition accuracy of the model for the training set load types. The proposed method has the highest recognition accuracy of 99.6%.

(5) And none of the three methods compared are evaluated for recognition ability on non-training set loads and interference test was carried out either. This is non-negligible evaluation factor that determines the generalization ability of the model. The proposed method considers load power and load types variation to evaluate the performance of the model for detection of non-training set loads.

## 6. Conclusion

Aiming at the problem of the low-voltage AC series arc fault has strong concealment and seriously threatens the safety of electricity use in civil applications. A series arc fault identification method based on based on wavelet transform and feature values decomposition fusion DNN is proposed. The main conclusions of this paper are as follows:




(a) Current generated by the process of putting in the microwave oven when the heater is in steady operation
(b) Graph of calculated results for the 3 parameter indicators
(c) Classification results of DNN fault detection model for load access circuit processes

**Fig. 12.** Analysis of the load access circuit process.

(1) The eigenvector $\sigma$ is extracted by wavelet transform and eigenvalue decomposition of the collected current signal. The three evaluation indexes $d_1$, $d_2$ and $d_3$ of $\sigma$ are formed into feature vector $d$. It is used as the driving data of DNN, and the arc feature information is effectively amplified. The complexity of feature information extraction is greatly simplified, and the input dimension of the model is reduced. With the benefits of traditional algorithms and neural networks, the training process is simplified.
(2) It is verified that the DNN fault detection model established in this paper has a good identification effect for both training set load and non-training set load types. The overall identification rate (training set load and non-training set load) of the arc half cycle in the fault current signal can reach 94.5%. GB/14,287.4 standard is satisfied and within its allowable range, and the overall recognition rate (training set load and non-training set load) of arc fault identification is around 98.7%. It provides analytical ideas for the identification of AC arc faults in low-voltage systems.

## CRediT authorship contribution statement

**Quanyi Gong:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Ke Peng:** Writing – review & editing. **Qun Gao:** Investigation, Visualization. **Liang Feng:** Software, Supervision. **Chuanliang Xiao:** Validation, Visualization.

**Table 6**
DNN model detection results of load startup process.

| Load name | The maximum number of misjudgments for the arc half cycle in 1s | The number of times the DNN model correctly identifies the startup state |
|---|---|---|
| Air conditioner 1200 W | 12 | 10 |
| Resistor 800 W | 8 | 10 |
| Electric iron 1000 W | 6 | 10 |
| Electric fan 400 W | 8 | 10 |
| Induction cooker 2100 W | 6 | 10 |
| Microwave oven 800 W | 2 | 10 |
| Vacuum cleaner 1000 W | 9 | 10 |
| Microwave oven 800 *W* + resistance 800 W (start at the same time) | 7 | 10 |
| Computer 600 *W* + Vacuum Cleaner 1 kW (start at the same time) | 9 | 10 |
| The resistance is stable and connected to the microwave oven | 8 | 10 |
| The computer is running steadily and connected to the vacuum cleaner | 9 | 10 |
| Fluorescent lamp 36W | 5 | 10 |

**Table 7**
Comparison of arc fault detection method comparison.

| Method | Reference [23] | Reference [30] | Reference [31] | Proposed method |
|---|---|---|---|---|
| Applicable load type range | 1,2,3,4 | 1,2,4,6,7 | 1,2,3,4,6 | 1–7 |
| Eigenvector scale | 1 × 500 | 1 × 20,000 | 5 × 2500 | 1 × 3 |
| Mean recognition time | Not introduced | 1 s | 0.5 s | 0.18 s |
| Overall recognition rate | 99.1% | Not introduced | 97.91% | 99.6% |
| Evaluate on non-training loads | Not evaluated | Not evaluated | Not evaluated | Recognition 90% |

Notes of applicable load type range: 1. Resistive 2. Capacitor start motor 3. Vortex 4. Switching power supply 5. Magnetron 6. Series motor 7. Combined load.

## Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Series arc fault identification method based on wavelet transform and feature values decomposition fusion DNN”.

## Data availability

Please contact the corresponding author if necessary for all data

## Acknowledgements

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